Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data.

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

ASTRONOMY & ASTROPHYSICS: HELIOPHYSICS WITH ADITYA L1 DATA

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PART 3

Independent Exploration of Solar Image Segmentation Using Aditya-L1 SUIT Data

Abstract

Solar image analysis plays an important role in understanding the structure and activity of the Sun. In this work, an Aditya-L1 SUIT FITS image was independently selected and processed using image processing techniques. The image was preprocessed through invalid value removal, intensity normalization, clipping, and logarithmic scaling. Three segmentation methods, namely Otsu Thresholding, Adaptive Thresholding, and K-Means Clustering, were applied and compared. A solar boundary visualization was also created as an additional analysis. The results show that Otsu Thresholding provides the most accurate boundary detection for the selected solar image.

Keywords: Aditya-L1, SUIT, FITS, Solar Image Processing, Otsu Thresholding, Adaptive Thresholding, K-Means Clustering

1.Introduction

The Sun continuously exhibits dynamic features that influence space weather and Earth's environment. The Aditya-L1 mission, launched by ISRO, carries the Solar Ultraviolet Imaging Telescope (SUIT) to capture high-resolution ultraviolet images of the Sun.

The objective of this work is to independently analyze a SUIT FITS image using different segmentation techniques and determine which method best identifies the solar region. The study also includes a creative visualization by overlaying the detected solar boundary on the original image.

2. Dataset Information

The dataset used in this work is a SUIT FITS image downloaded from the Aditya-L1 data archive.

Dataset Details

- Mission: Aditya-L1
- Instrument: SUIT (Solar Ultraviolet Imaging Telescope)
- File Format: FITS
- Selected File:

  `SUT_T26_0723_002181_Lev1.0_2026-05-25T21.09.53.750_0973NB07.fits`

The selected image was chosen because it contains a clearly visible solar limb suitable for segmentation and boundary detection.

3. Research Question

Which image segmentation technique provides the most effective detection of the solar region in the selected Aditya-L1 SUIT image?

4. Methodology

The following workflow was used for image analysis.

1. Load FITS file.
2. Extract image data.
3. Remove invalid pixel values.
4. Perform intensity clipping.
5. Apply logarithmic scaling.
6. Normalize image intensity.
7. Apply Otsu Thresholding.
8. Apply Adaptive Thresholding.
9. Apply K-Means Clustering.
10. Compare segmentation methods.
11. Visualize detected solar boundary.

5. Image Preprocessing

The FITS image was preprocessed before segmentation.

The preprocessing operations included:

- Removal of invalid values
- Intensity clipping
- Logarithmic scaling
- Intensity normalization

These operations improved the visibility of the solar image and prepared it for segmentation.

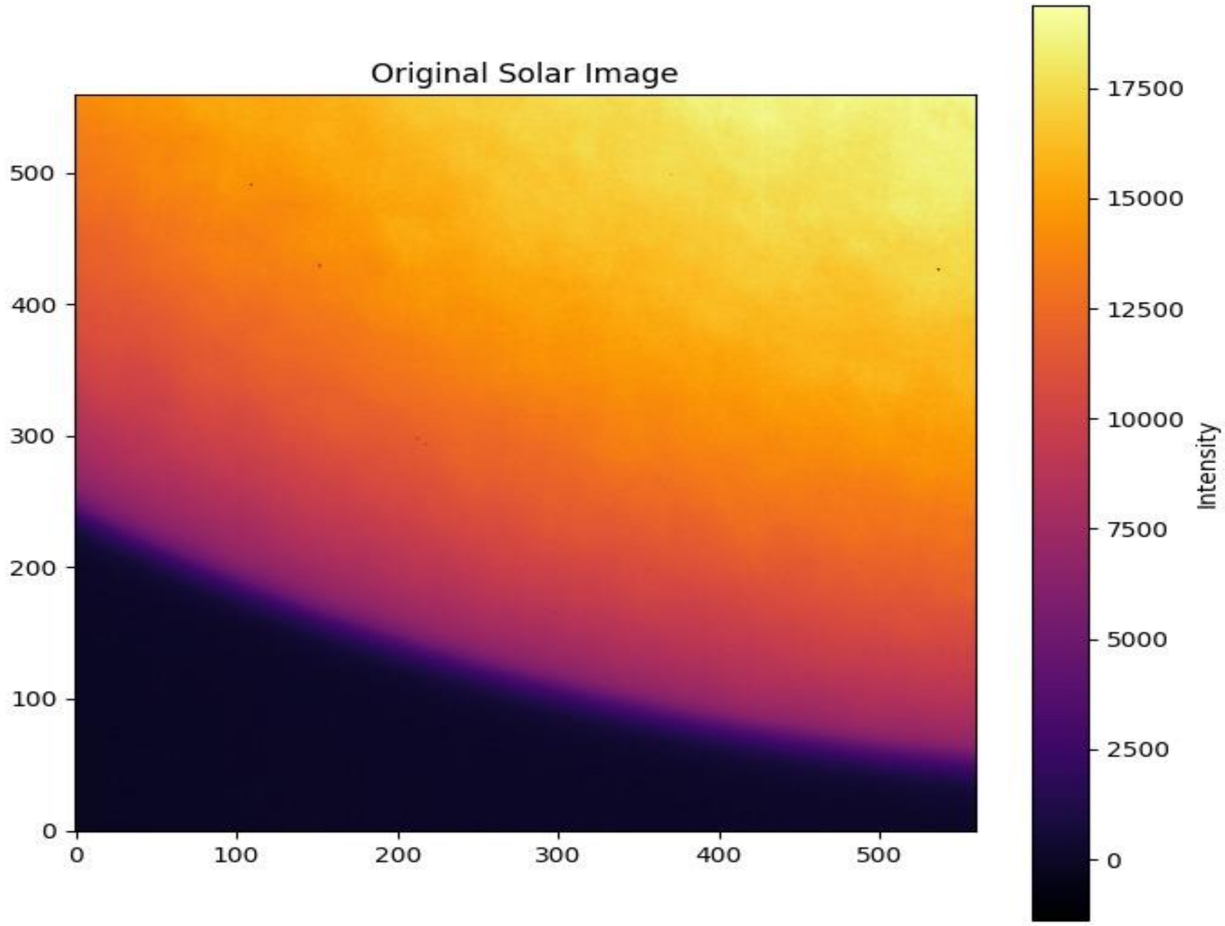

Figure 1. Original Solar Image

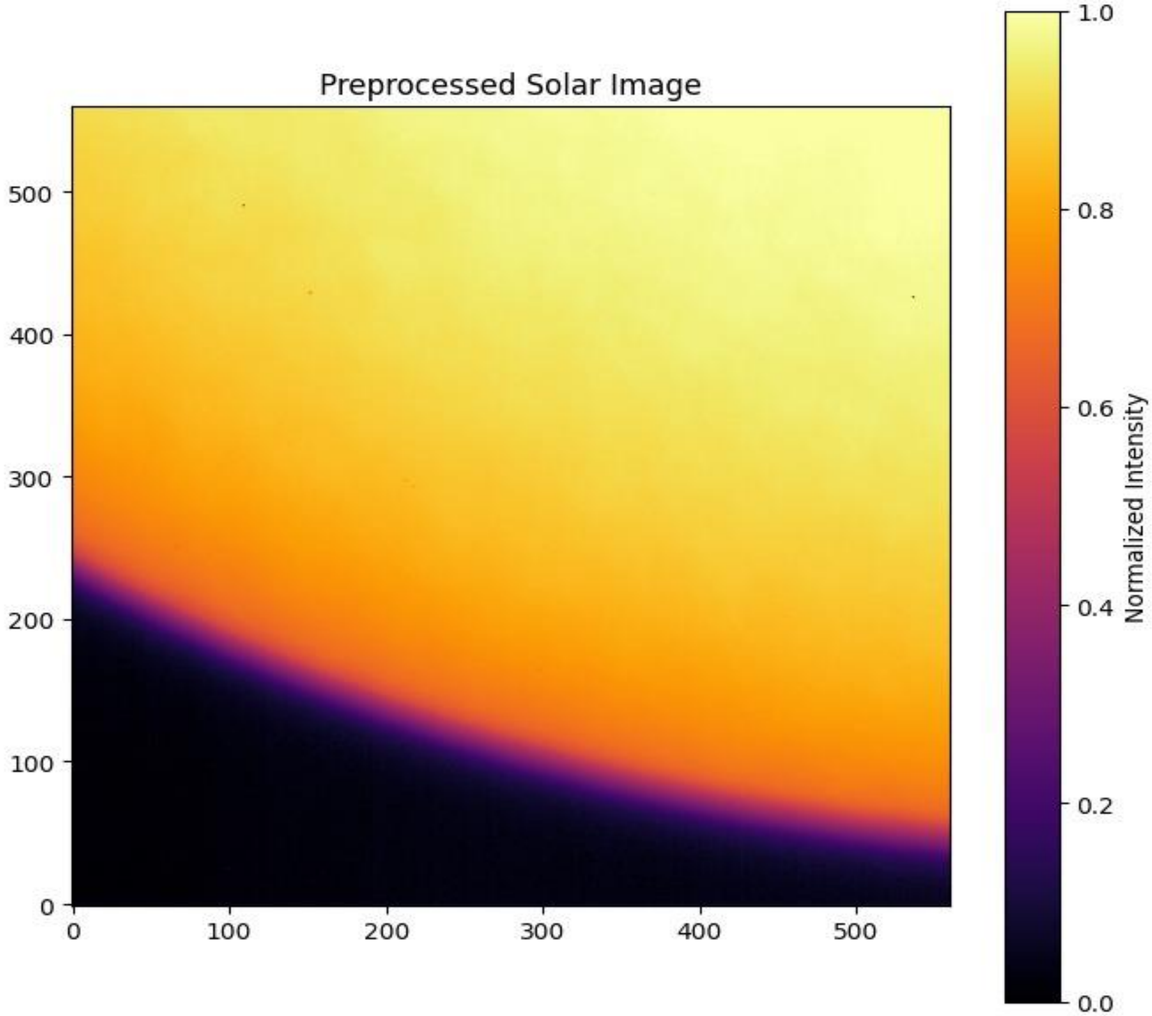

Figure 2. Preprocessed Solar Image

6. Image Segmentation

6.1 Otsu Thresholding

Otsu Thresholding automatically determines a global threshold that separates the solar disk from the background. The method produced a clean segmentation with a clear solar boundary.

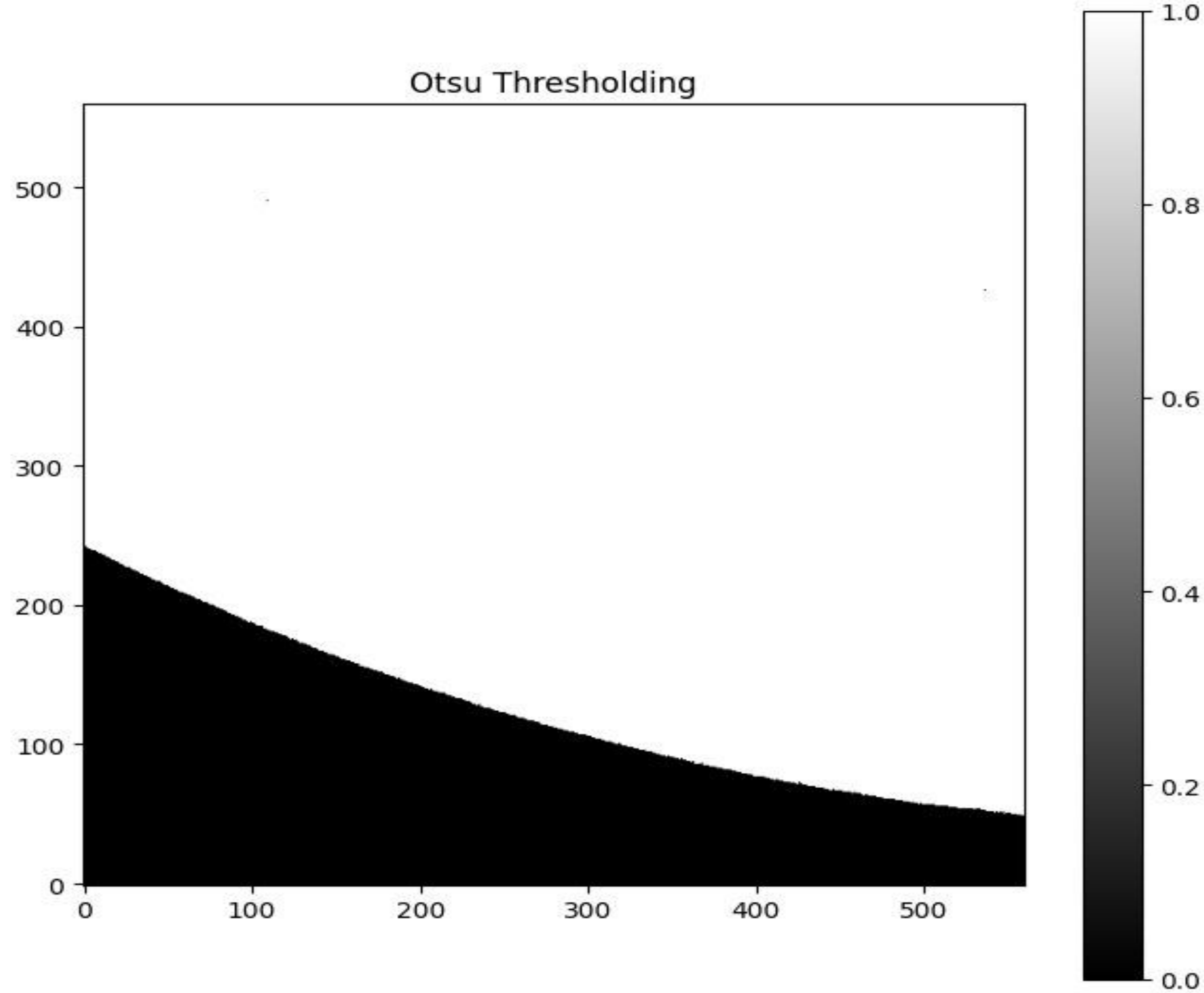

Figure 3. Otsu Thresholding Result

6.2 Adaptive Thresholding

Adaptive Thresholding calculates local thresholds for different regions of the image. It detected local intensity variations but introduced additional noise because of the gradual brightness changes in the image.

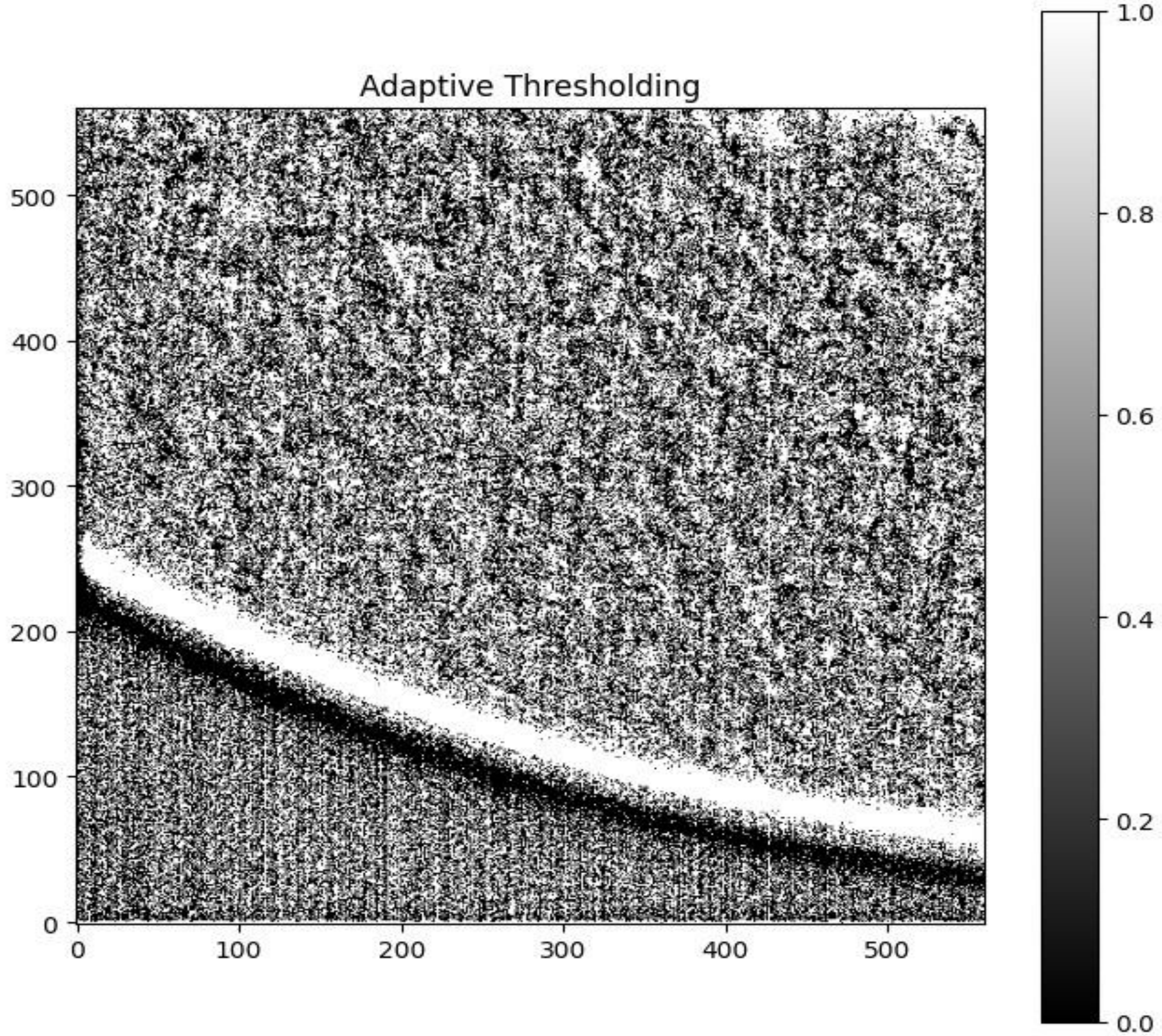

Figure 4. Adaptive Thresholding Result

6.3 K-Means Clustering

K-Means Clustering grouped image pixels into two clusters according to their intensity values. The segmented image was similar to the Otsu result and successfully separated the solar region from the background.

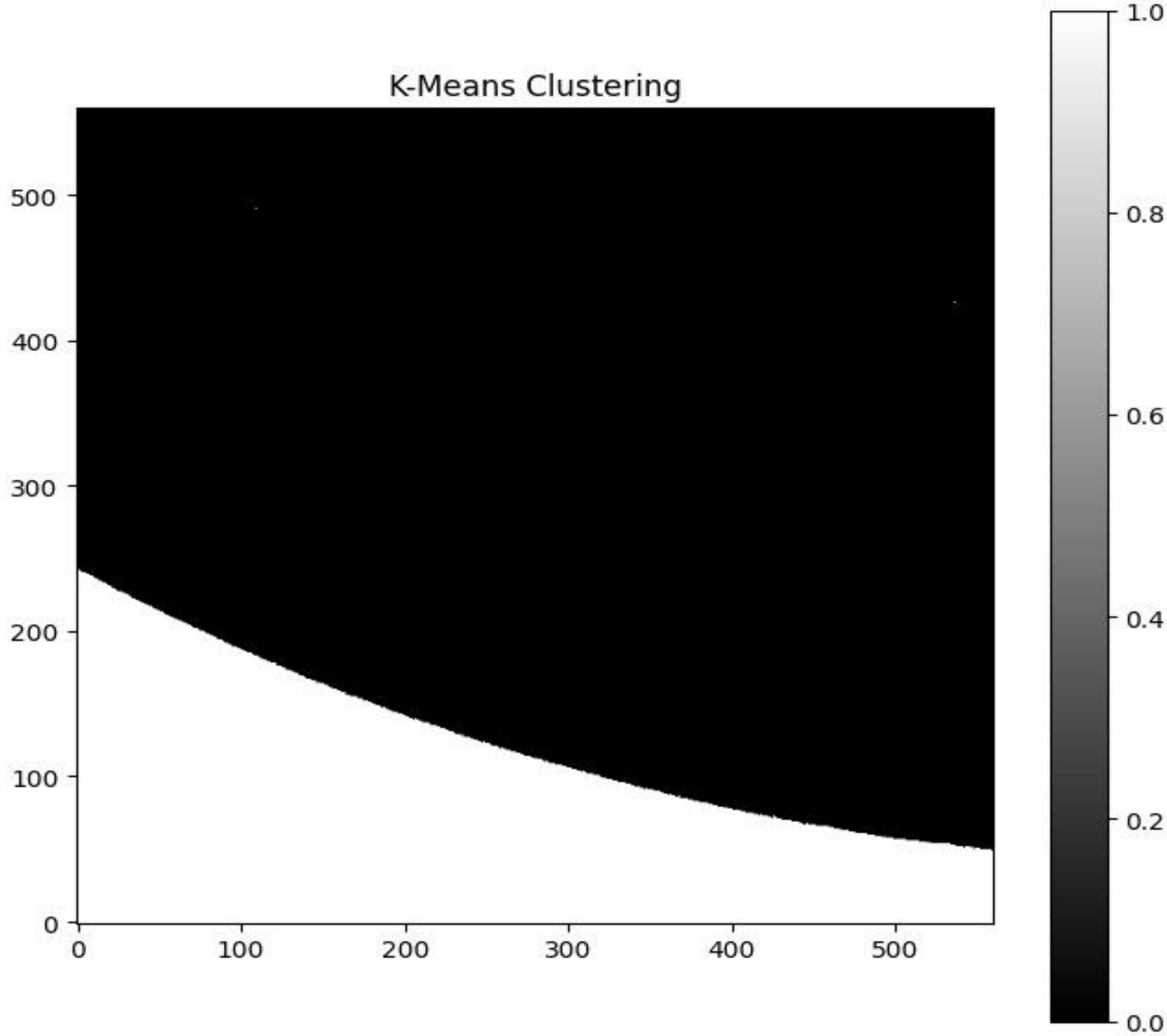

Figure 5. K-Means Clustering Result

7. Comparison of Segmentation Methods

The performance of the three segmentation methods was compared by analyzing the detected foreground pixels.

- Otsu Thresholding produced a clean global segmentation.
- Adaptive Thresholding detected more local variations but also introduced noise.
- K-Means Clustering generated a segmentation comparable to Otsu Thresholding using unsupervised clustering.

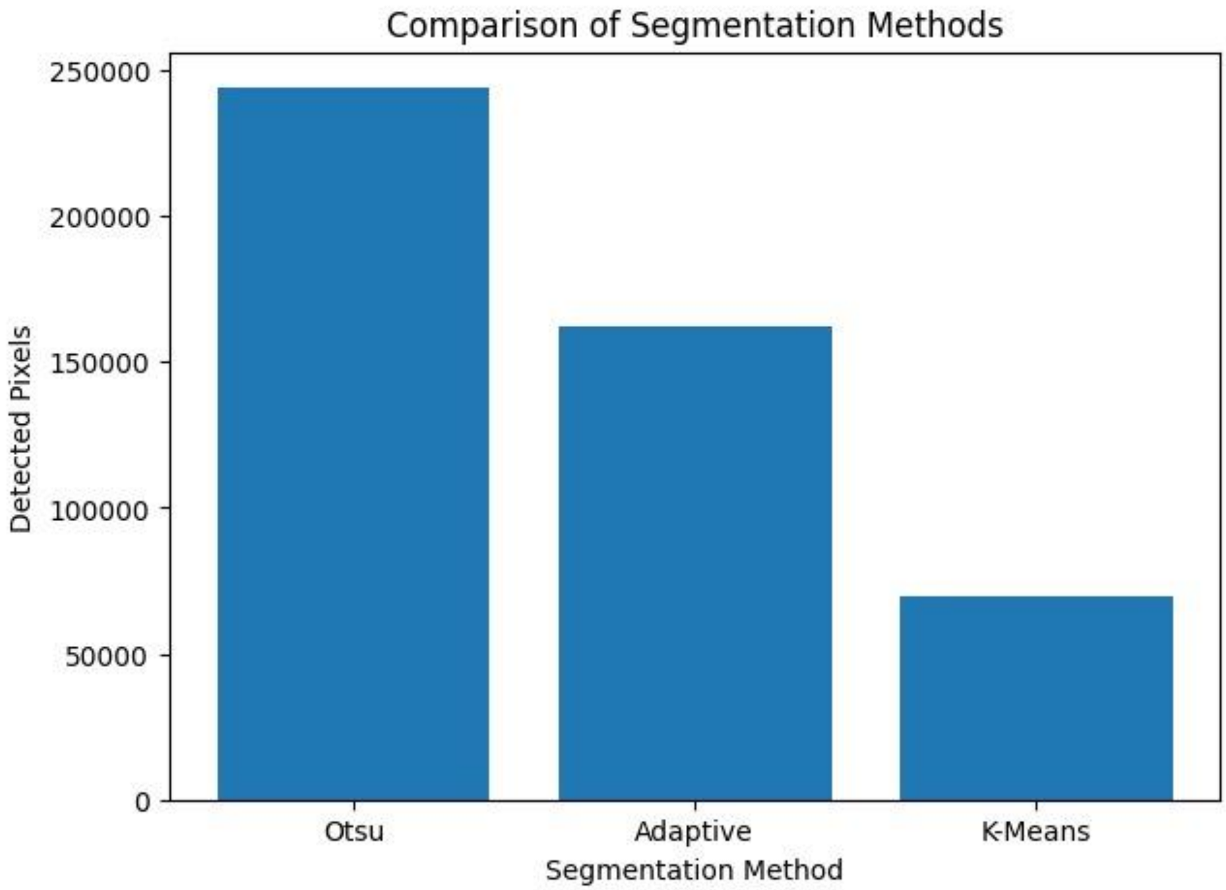

Figure 6. Comparison of Segmentation Methods

8. Creative Visualization

To improve the interpretation of the segmentation results, the detected solar boundary obtained using Otsu Thresholding was overlaid on the original solar image using a cyan contour.

The boundary closely followed the visible solar limb, confirming the accuracy of the segmentation.

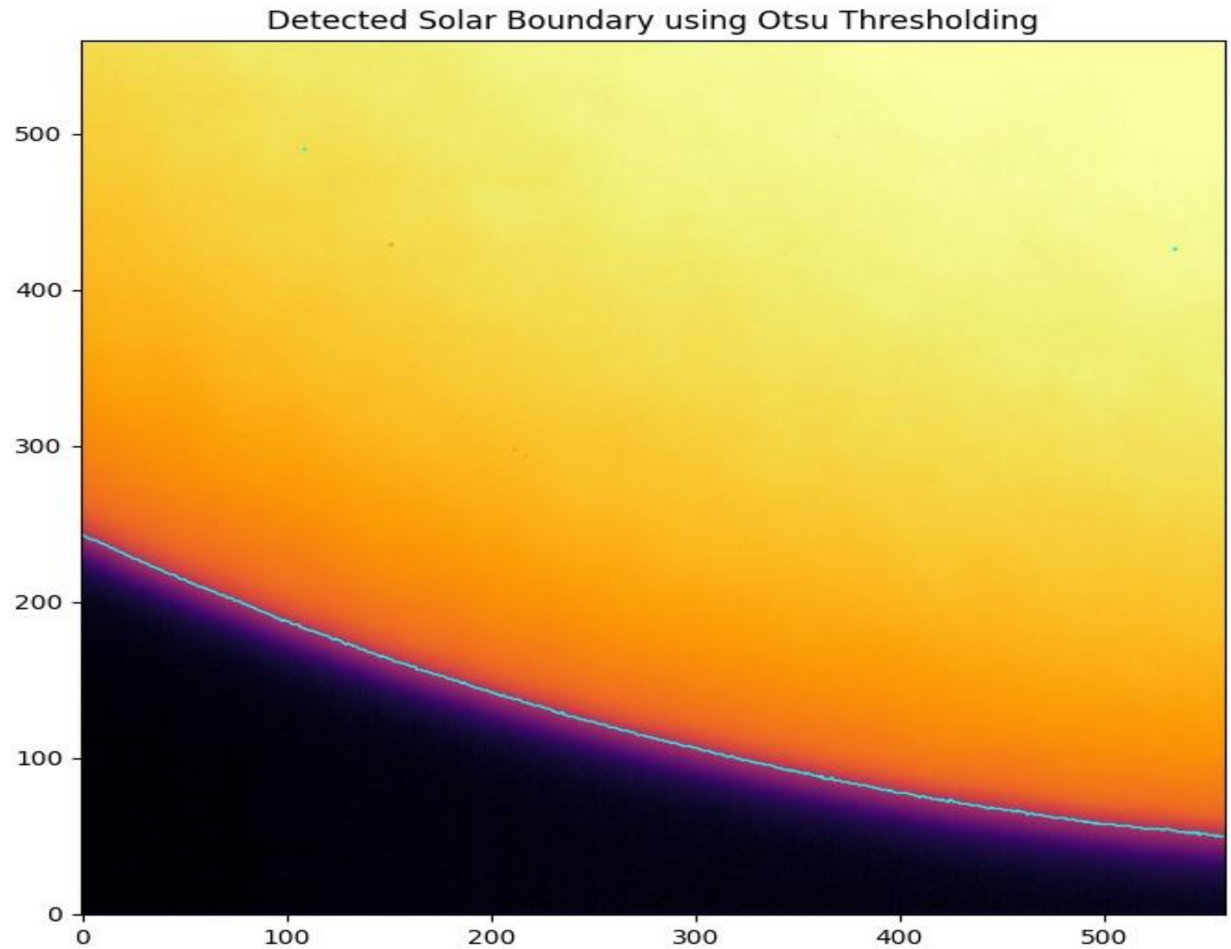

Figure 7. Detected Solar Boundary Using Otsu Thresholding

9. Results and Discussion

The selected Aditya-L1 SUIT image was successfully analyzed using image preprocessing and segmentation techniques. Image preprocessing enhanced the visibility of the solar image and prepared it for further analysis.

Otsu Thresholding accurately separated the solar disk from the background using a global threshold. Adaptive Thresholding detected local intensity variations but introduced noticeable noise. K-Means Clustering grouped pixels according to their intensity values and produced a segmentation similar to Otsu Thresholding.

The comparison of the three methods showed that Otsu Thresholding provided the most accurate and clean segmentation for the selected image. The creative boundary visualization further confirmed that the detected contour accurately represented the solar limb.

10. Conclusion

This study demonstrated the application of image processing techniques for analyzing Aditya-L1 SUIT solar images. The selected FITS image was successfully preprocessed and segmented using Otsu Thresholding, Adaptive Thresholding, and K-Means Clustering.

Among the three techniques, Otsu Thresholding provided the best segmentation result by accurately detecting the solar boundary with minimal noise. K-Means Clustering also produced satisfactory results, while Adaptive Thresholding was more sensitive to local intensity variations and noise.

The boundary visualization improved the interpretation of the segmentation results and validated the effectiveness of the selected method. This work demonstrates that basic image processing techniques can effectively analyze solar images and can be extended for future solar feature detection studies.

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